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WO-300-01 — Pull, clean and test the tube bundle of E-101

Area	A300 Exchangers and drum	**Task**	XB Exchanger bundle pull, clean and tube-side test
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**E-101** — E-101 shell-and-tube exchanger 36" ID × 192" — E 111'-6" N -25'-0" EL 106'-6"		
Equipment	E-101	**Lines**	4"-P300-P-1003, 4"-P300-P-1005, 4"-P150-CWS-3001, 4"-P150-CWR-3002, 1"-P150-V-6021, 1"-P150-D-6022, 3/4"-P300-I-6023
Isometrics	ISO-4"-P300-P-1003, ISO-4"-P300-P-1005, ISO-4"-P150-CWS-3001, ISO-4"-P150-CWR-3002, ISO-1"-P150-V-6021, ISO-1"-P150-D-6022, ISO-3/4"-P300-I-6023		
Permits	General work permit, Confined-space entry permit, Lifting plan		

Reason

Fouling: tube bundle to be pulled, cleaned and tested.

Scope of work

1. Isolate E-101 per the isolation list (tube and shell sides); drain and vent.
2. Remove the channel cover; pull the tube bundle to the cleaning bay.
3. Clean the tubes and shell side; inspect visually; thickness at the TMLs.
4. Reinstall the bundle with new gaskets; tube-side hydrostatic test at 420 psi (SW-TEST-1).
5. Remove blinds; restore the line-up.

Affected items: E-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-101	equipment	E-101 shell-and-tube exchanger 36" ID × 192"		E 111'-6" N -25'-0" EL 106'-6"
4"-P300-P-1003/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1003	E 110'-6" N -25'-0" EL 109'-3-13/16"
E-101-N1	nozzle	Nozzle E-101-N1 (tube in, 4" P300) on E-101		E 110'-6" N -25'-0" EL 108'-8-1/16"
4"-P300-P-1005/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 110'-6" N -25'-0" EL 103'-8-3/16"
E-101-N2	nozzle	Nozzle E-101-N2 (tube out, 4" P300) on E-101		E 110'-6" N -25'-0" EL 104'-3-15/16"
4"-P150-CWS-3001/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3001	E 113'-6" N -25'-0" EL 109'-3-13/16"
E-101-N3	nozzle	Nozzle E-101-N3 (shell in, 4" P150) on E-101		E 113'-6" N -25'-0" EL 108'-8-1/16"
4"-P150-CWR-3002/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWR-3002	E 125'-6" N -25'-0" EL 103'-8-3/16"
E-101-N4	nozzle	Nozzle E-101-N4 (shell out, 4" P150) on E-101		E 125'-6" N -25'-0" EL 104'-3-15/16"
1"-P150-V-6021/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6021	E 124'-6" N -25'-0" EL 109'-1/2"
E-101-N5	nozzle	Nozzle E-101-N5 (vent, 1" P150) on E-101		E 124'-6" N -25'-0" EL 108'-5-11/16"
1"-P150-D-6022/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6022	E 113'-6" N -25'-0" EL 103'-11-1/2"
E-101-N6	nozzle	Nozzle E-101-N6 (drain, 1" P150) on E-101		E 113'-6" N -25'-0" EL 104'-6-3/8"
3/4"-P300-I-6023/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6023	E 119'-6" N -27'-6-1/4" EL 106'-6"
E-101-N7	nozzle	Nozzle E-101-N7 (instrument, 0.75" P300) on E-101		E 119'-6" N -26'-11-1/2" EL 106'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1003/B2	4"-P300-P-1003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1003/B2) on E-101-N1	E 110'-6" N -25'-0" EL 109'-3-13/16"		
2	blind	E-101-N1	4"-P300-P-1003	Fit a blind between 4"-P300-P-1003/B2 and the nozzle flange E-101-N1	E 110'-6" N -25'-0" EL 108'-8-1/16"		
3	valve	4"-P300-P-1005/A2	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/A2) on E-101-N2	E 110'-6" N -25'-0" EL 103'-8-3/16"		
4	blind	E-101-N2	4"-P300-P-1005	Fit a blind between 4"-P300-P-1005/A2 and the nozzle flange E-101-N2	E 110'-6" N -25'-0" EL 104'-3-15/16"		

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
5	valve	4"-P150-CWS-3001/B2	4"-P150-CWS-3001	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3001/B2) on E-101-N3	E 113'-6" N -25'-0" EL 109'-3-13/16"		
6	blind	E-101-N3	4"-P150-CWS-3001	Fit a blind between 4"-P150-CWS-3001/B2 and the nozzle flange E-101-N3	E 113'-6" N -25'-0" EL 108'-8-1/16"		
7	valve	4"-P150-CWR-3002/A2	4"-P150-CWR-3002	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWR-3002/A2) on E-101-N4	E 125'-6" N -25'-0" EL 103'-8-3/16"		
8	blind	E-101-N4	4"-P150-CWR-3002	Fit a blind between 4"-P150-CWR-3002/A2 and the nozzle flange E-101-N4	E 125'-6" N -25'-0" EL 104'-3-15/16"		
9	valve	1"-P150-V-6021/2	1"-P150-V-6021	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6021/2) on E-101-N5	E 124'-6" N -25'-0" EL 109'-1/2"		
10	blind	E-101-N5	1"-P150-V-6021	Fit a blind between 1"-P150-V-6021/2 and the nozzle flange E-101-N5	E 124'-6" N -25'-0" EL 108'-5-11/16"		
11	valve	1"-P150-D-6022/2	1"-P150-D-6022	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6022/2) on E-101-N6	E 113'-6" N -25'-0" EL 103'-11-1/2"		
12	blind	E-101-N6	1"-P150-D-6022	Fit a blind between 1"-P150-D-6022/2 and the nozzle flange E-101-N6	E 113'-6" N -25'-0" EL 104'-6-3/8"		
13	valve	3/4"-P300-I-6023/2	3/4"-P300-I-6023	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6023/2) on E-101-N7	E 119'-6" N -27'-6-1/4" EL 106'-6"		
14	blind	E-101-N7	3/4"-P300-I-6023	Fit a blind between 3/4"-P300-I-6023/2 and the nozzle flange E-101-N7	E 119'-6" N -26'-11-1/2" EL 106'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-300-01-I1	Internal visual examination of the shell, channel and tube bundle	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-300-01-I2	Ultrasonic thickness at the 12 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-01-TML01	E-101	Nozzle neck E-101-N1	E 110'-6" N -25'-0" EL 108'-8-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML02	E-101	Nozzle neck E-101-N2	E 110'-6" N -25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML03	E-101	Nozzle neck E-101-N3	E 113'-6" N -25'-0" EL 108'-8-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML04	E-101	Nozzle neck E-101-N4	E 125'-6" N -25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML05	E-101	Nozzle neck E-101-N5	E 124'-6" N -25'-0" EL 108'-5-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML06	E-101	Nozzle neck E-101-N6	E 113'-6" N -25'-0" EL 104'-6-3/8"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML07	E-101	Nozzle neck E-101-N7	E 119'-6" N -26'-11-1/2" EL 106'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML08	E-101	Left head centre	E 110'-8-3/4" N -25'-0" EL 106'-6"		centre: ____			
WO-300-01-TML09	E-101	Right head centre	E 128'-3-1/4" N -25'-0" EL 106'-6"		centre: ____			
WO-300-01-TML10	E-101	Shell at 0.000" from the left tangent	E 111'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-01-TML11	E-101	Shell at 6'-0" from the left tangent	E 117'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-01-TML12	E-101	Shell at 12'-0" from the left tangent	E 123'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Hydrostatic test (tube side)

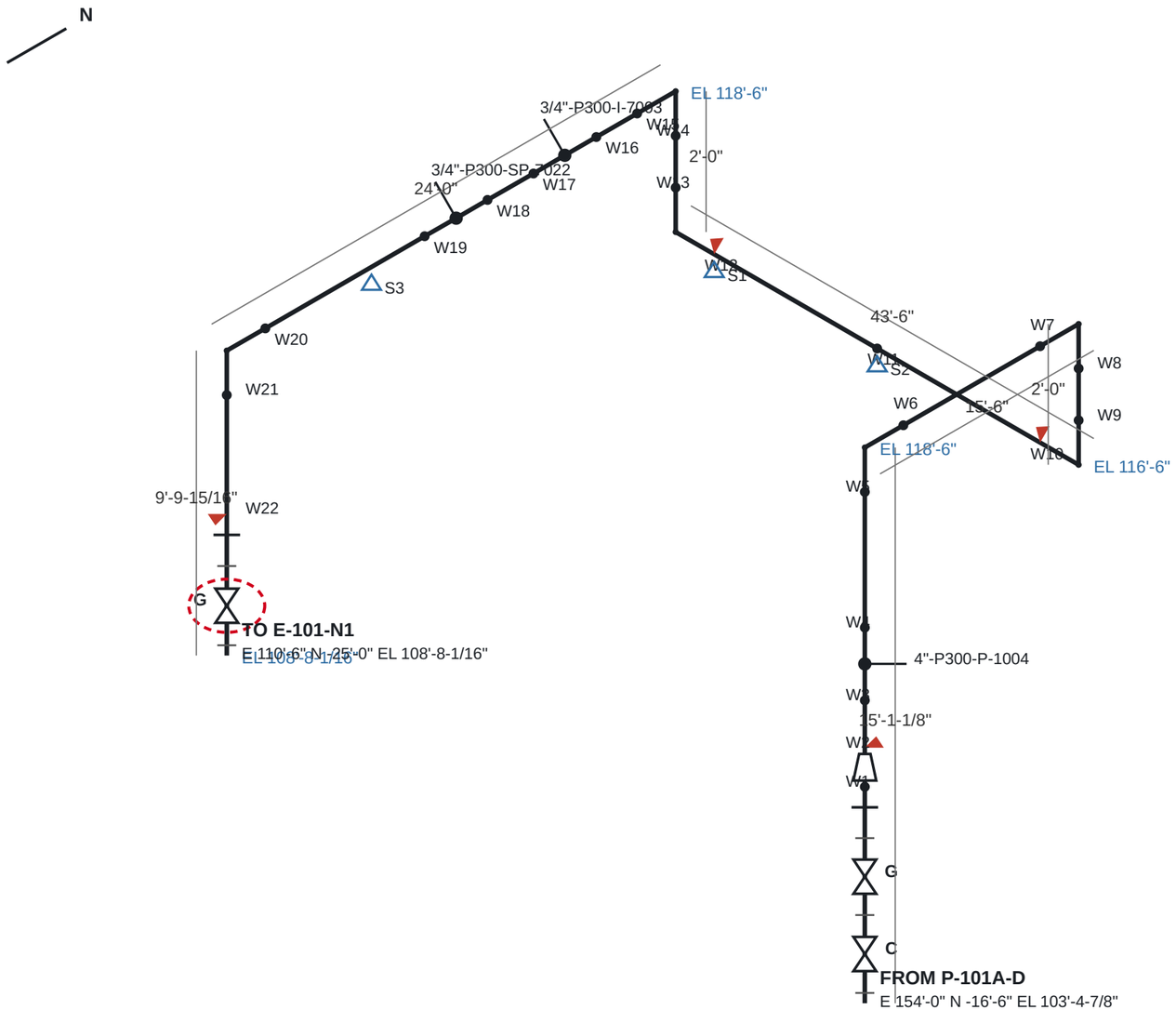
Description	Tube side of the reinstalled bundle
Pressure	420 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Channel cover removed; bundle pulled and cleaned	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 12 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Tube-side hydrostatic test at 420 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record	H	H	H				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1003 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	REDUCER CONCENTRIC 4" × 3" P300	1
7	PIPE 4" P300 BEVELLED ENDS	98'-8-7/16"
8	TEE EQUAL 4" P300	1
9	ELBOW 90° R = 5/3 OD 4" P300	6
10	TEE REDUCING 4" × 3/4" P300	2
11	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	2
12	FLANGE WELD-NECK 4" P300 RAISED FACE	1
13	GASKET 4" P300 COMPRESSED SHEET	2
14	VALVE, GATE, WEDGE, OS&Y 4" P300	1

WELDS 22 (4 FIELD) · SUPPORTS 3 · SPOOLS 5

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12F W13 W14 W15 W16 W17 W18 W19 W20
W21 W22F

4"-P300-P-1003

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-101A-D → E-101-N1
PIPE 98'-8-7/16" · WEIGHT 1276 lb

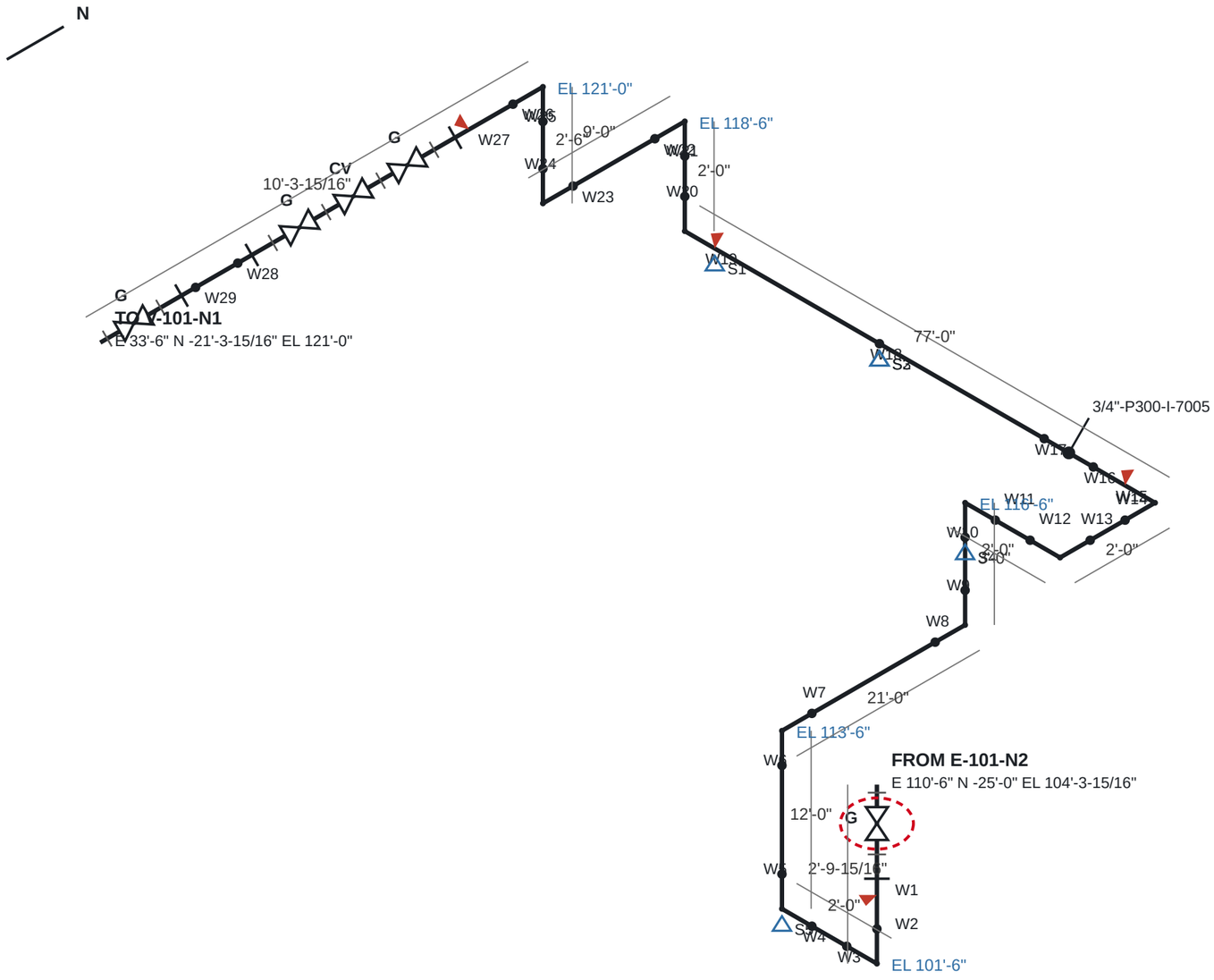
DWG ISO-4"-P300-P-1003 REV 0 WO-300-01

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Isometric ISO-4"-P300-P-1005

ISOMETRIC — 4"-P300-P-1005 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	GASKET 4" P300 COMPRESSED SHEET	8
2	VALVE, GATE, WEDGE, OS&Y 4" P300	4
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	8
4	FLANGE WELD-NECK 4" P300 RAISED FACE	4
5	PIPE 4" P300 BEVELLED ENDS	124'-6-1/2"
6	ELBOW 90° R = 5/3 OD 4" P300	11
7	TEE REDUCING 4" × 3/4" P300	1
8	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1

WELDS 29 (4 FIELD) · SUPPORTS 5 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F W16 W17 W18 W19F W20
W21 W22 W23 W24 W25 W26 W27F W28 W29

4"-P300-P-1005

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
E-101-N2 → V-101-N1
PIPE 124'-6-1/2" · WEIGHT 1974 lb

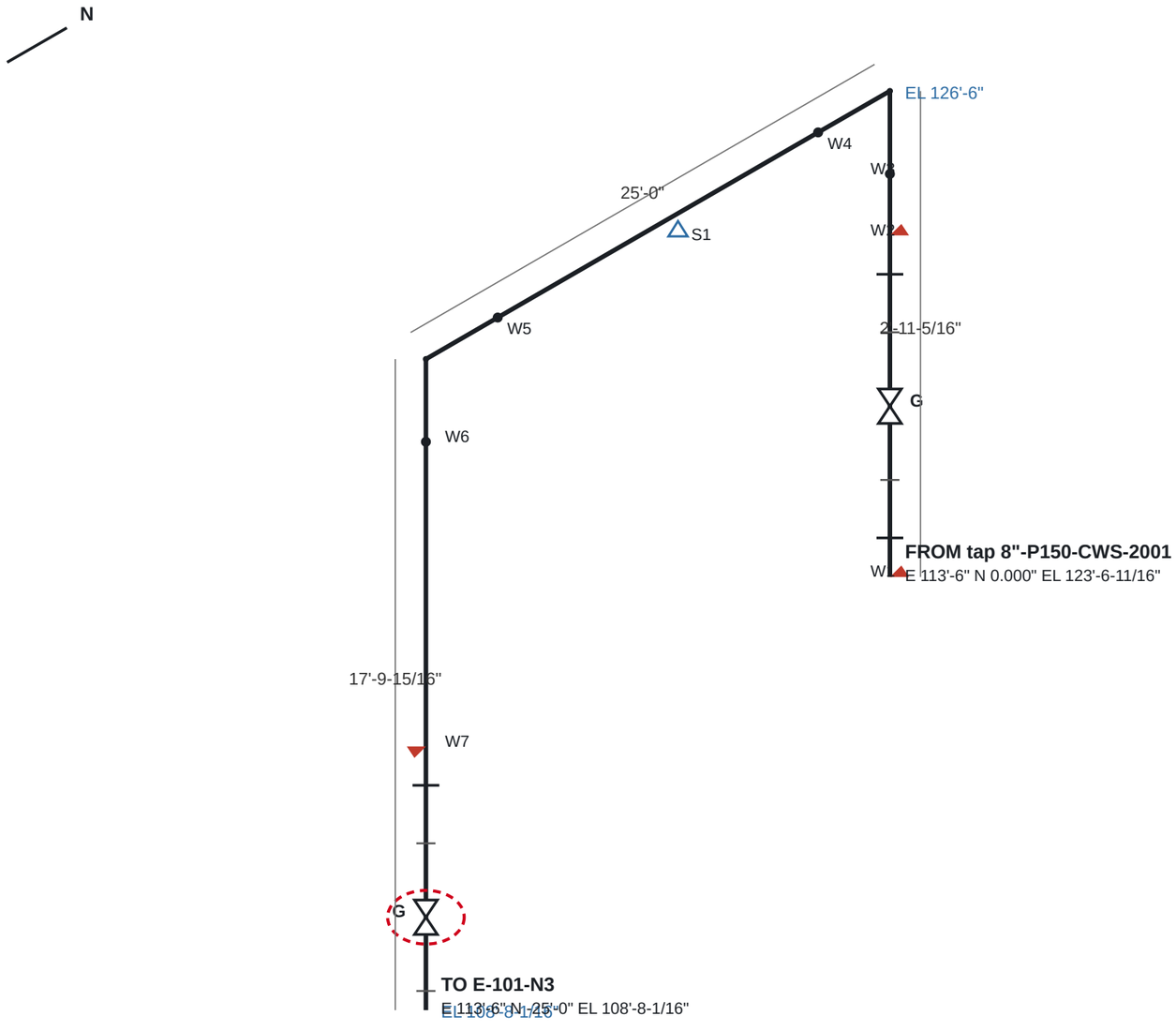
DWG ISO-4"-P300-P-1005 REV 0 WO-300-01

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Isometric ISO-4"-P150-CWS-3001

ISOMETRIC — 4"-P150-CWS-3001 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	FLANGE WELD-NECK 4" P150 RAISED FACE	3
2	GASKET 4" P150 COMPRESSED SHEET	4
3	VALVE, GATE, WEDGE, OS&Y 4" P150	2
4	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	4
5	PIPE 4" P150 BEVELLED ENDS	39'-7-5/16"
6	ELBOW 90° R = 5/3 OD 4" P150	2

WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

4"-P150-CWS-3001

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-101-N3
PIPE 39'-7-5/16" · WEIGHT 693 lb

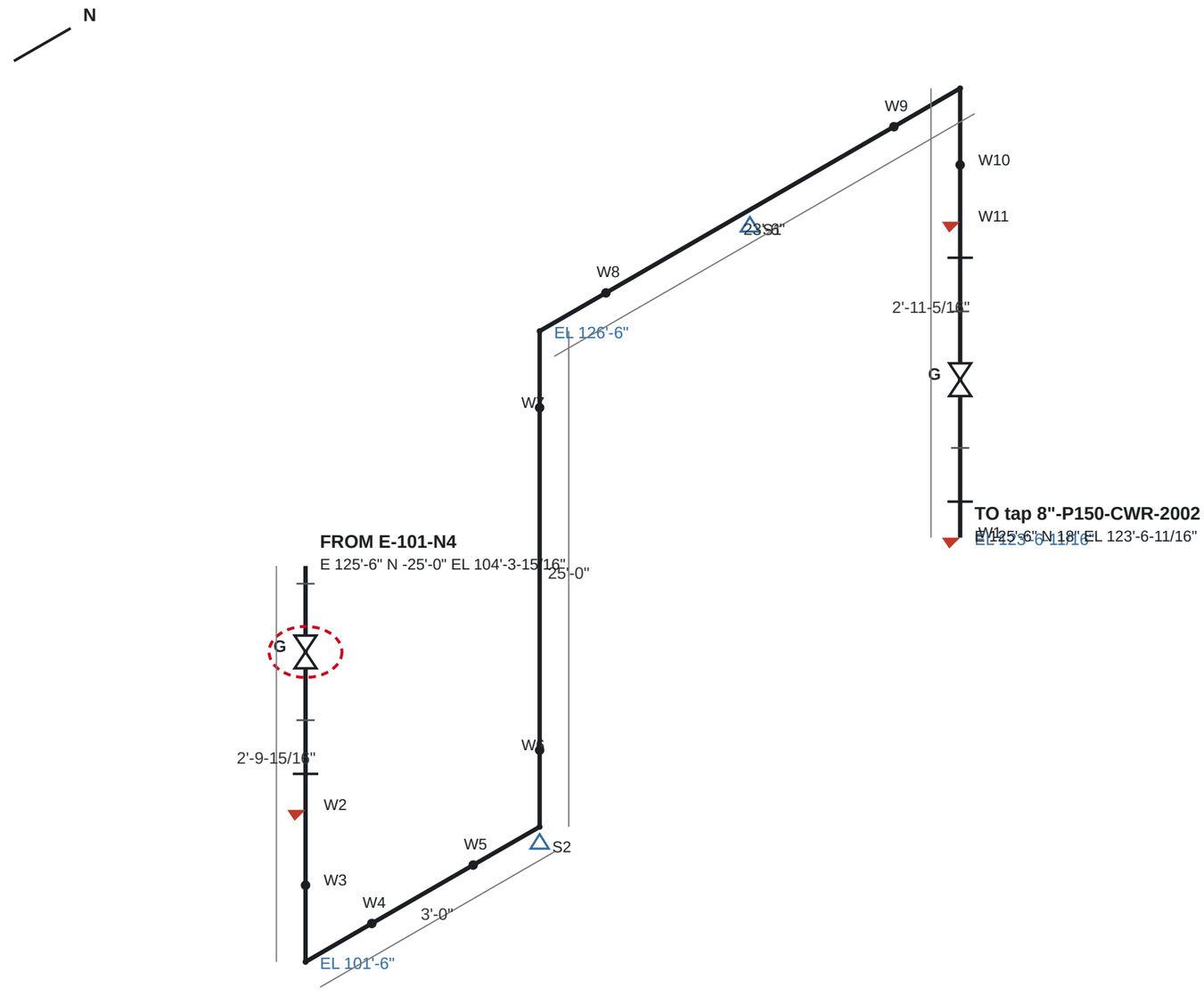
DWG ISO-4"-P150-CWS-3001 REV 0 WO-300-01

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Isometric ISO-4"-P150-CWR-3002

ISOMETRIC — 4"-P150-CWR-3002 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	GASKET 4" P150 COMPRESSED SHEET	4
2	VALVE, GATE, WEDGE, OS&Y 4" P150	2
3	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	4
4	FLANGE WELD-NECK 4" P150 RAISED FACE	3
5	PIPE 4" P150 BEVELLED ENDS	48'-10-5/8"
6	ELBOW 90° R = 5/3 OD 4" P150	4

WELDS 11 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11F

4"-P150-CWR-3002

CLASS W150 — Water (cooling and process)
SERVICE CWR · 4" · P150 · DESIGN 150 psi
E-101-N4 → tap 8"-P150-CWR-2002
PIPE 48'-10-5/8" · WEIGHT 777 lb

DWG ISO-4"-P150-CWR-3002 REV 0 WO-300-01

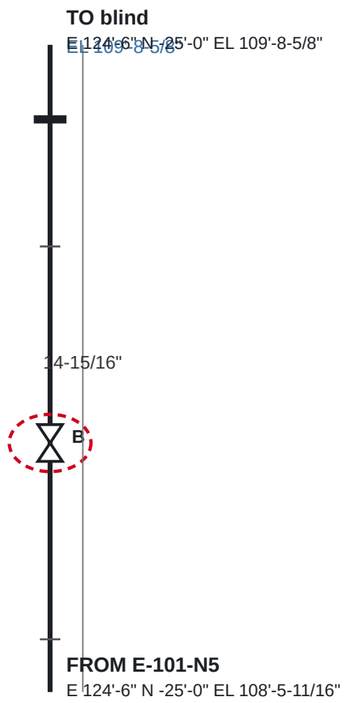
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Isometric ISO-1"-P150-V-6021

ISOMETRIC — 1"-P150-V-6021 — Pull, clean and test the tube bundle of E-101

N



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-V-6021

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
E-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

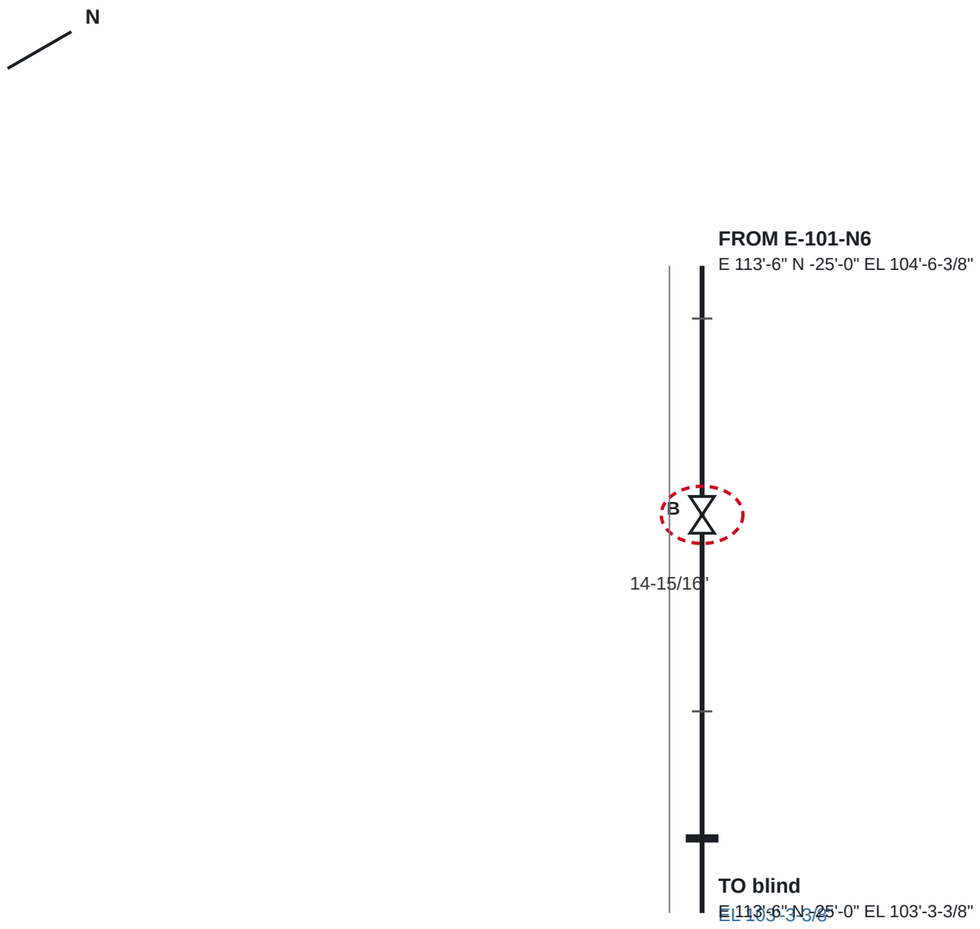
DWG ISO-1"-P150-V-6021 REV 0 WO-300-01

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Isometric ISO-1"-P150-D-6022

ISOMETRIC — 1"-P150-D-6022 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	GASKET 1" P150 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 1" P150	1
3	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150	2
4	FLANGE BLIND 1" P150 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

1"-P150-D-6022

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
E-101-N6 → blind
PIPE 0.000" · WEIGHT 39 lb

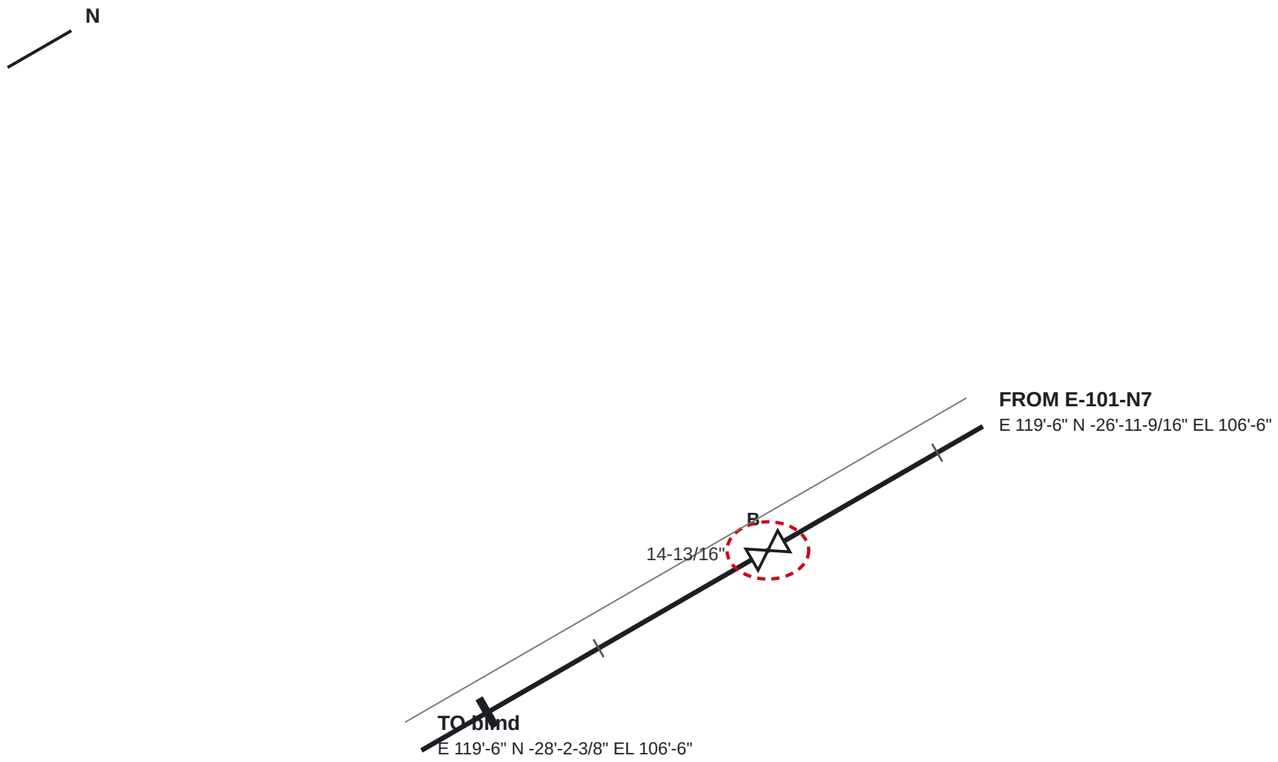
DWG ISO-1"-P150-D-6022 REV 0 WO-300-01

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Isometric ISO-3/4"-P300-I-6023

ISOMETRIC — 3/4"-P300-I-6023 — Pull, clean and test the tube bundle of E-101



BILL OF MATERIALS

1	GASKET 3/4" P300 COMPRESSED SHEET	2
2	VALVE, BALL, FLOATING 3/4" P300	1
3	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300	2
4	FLANGE BLIND 3/4" P300 RAISED FACE	1

WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0

3/4"-P300-I-6023

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
E-101-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

DWG ISO-3/4"-P300-I-6023 REV 0 WO-300-01

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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